

Question:
At birth, the least developed part of the brain is the

- Answer Choices:
- A. visual system
 - B. thalamus
 - C. frontal lobe
 - D. brain stem
 - E. parietal lobe
 - F. cerebellum
 - G. cortex
 - H. basal ganglia
 - I. hypothalamus
 - J. limbic system

Reasoning + Answer:

</tool_call> reasoning process here

At birth , the brain undergoes rapid development , with certain regions developing earlier than others . The brain stem , which controls vital functions like breathing and heart rate , is present and functional from birth . The thalamus , hypothalamus , and brain stem are among the earliest structures to develop and are essential for basic survival functions .

In contrast , the cerebral cortex , especially the prefrontal cortex (part of the frontal lobe) , is the last to mature . The frontal lobe is responsible for higher cognitive functions such as decision-making , planning , and executive control , and it develops gradually throughout childhood and adolescence . While the cortex is present at birth , its full development occurs over time , with the most advanced and complex regions like the prefrontal cortex maturing last .

Therefore , the least developed part of the brain at birth , in terms of structural and functional maturation , is the frontal lobe , particularly its cortical regions .

Other structures like the cerebellum and limbic system develop progressively but earlier than the prefrontal cortex .

Thus , the correct answer is the frontal lobe , as it is the last to develop structurally and functionally .

< answer > C < / answer >

